Use of PETase for PET Degradation in a Bioreactor to Improve Effectivity of Recycling

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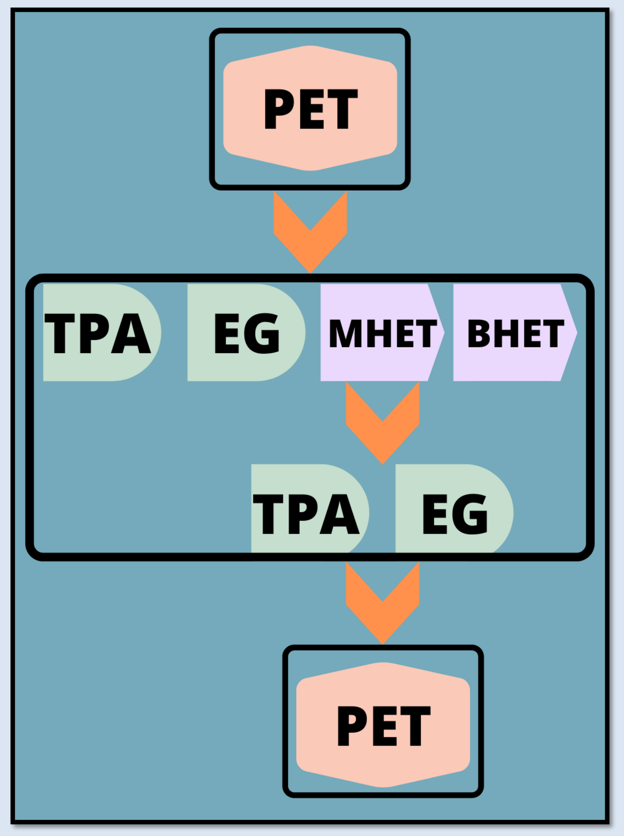
Graphical Abstract



With this technology, an input of PET plastic will result in an output of its building blocks, which can be remade into the plastic to create a more effective recycling system.

Background

Plastic Pollution

Plastic pollution is growing at an unsustainable rate, and these polymers do not naturally decay well. PET is one common type of polymer that is used in bottles and clothing.

Recycling

No real effective method of recycling is being implemented, which is detrimental in a consumer-based society, and currently it is more financially beneficial for manufacturers to make new plastic rather than reusing plastic. In under-developed countries, economies are not stable enough to support waste-management systems. Existing ways of disposing of plastic harm the environment.

Enzymes

Ideonella sakaiensis 201-F6 has developed an enzyme that aids in the process of breaking down PET and PEF for carbon and energy. Recent studies have looked at modifying this enzyme to better function for industrial use.

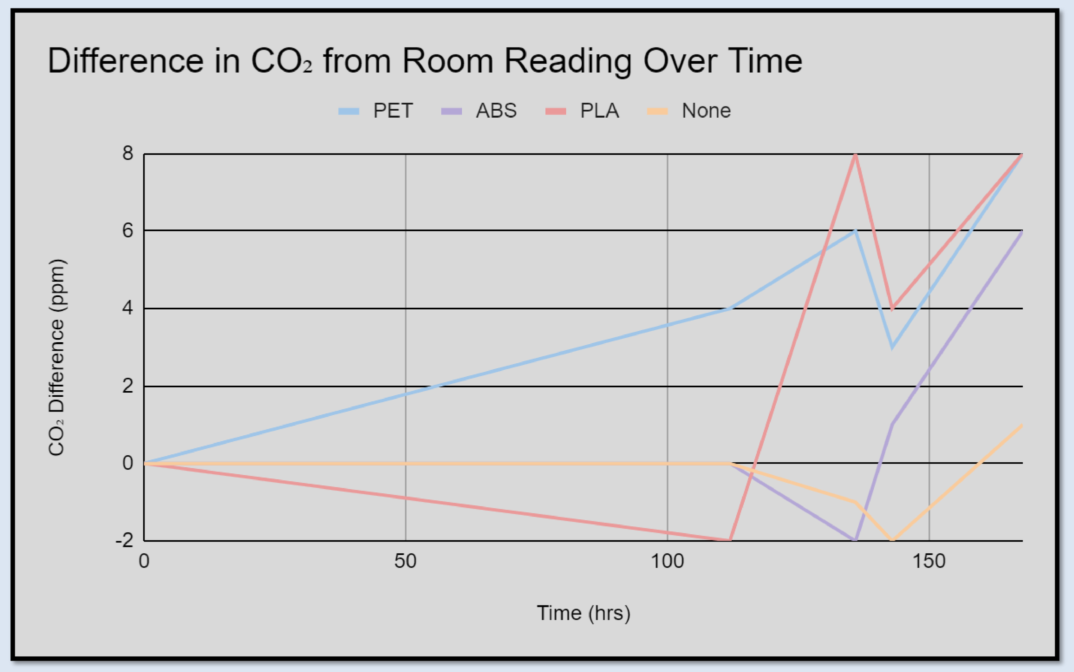
Engineering Need

As industries continue to mass produce plastic, it is becoming more apparent that there is no effective way to recycle plastic and address plastic pollution.

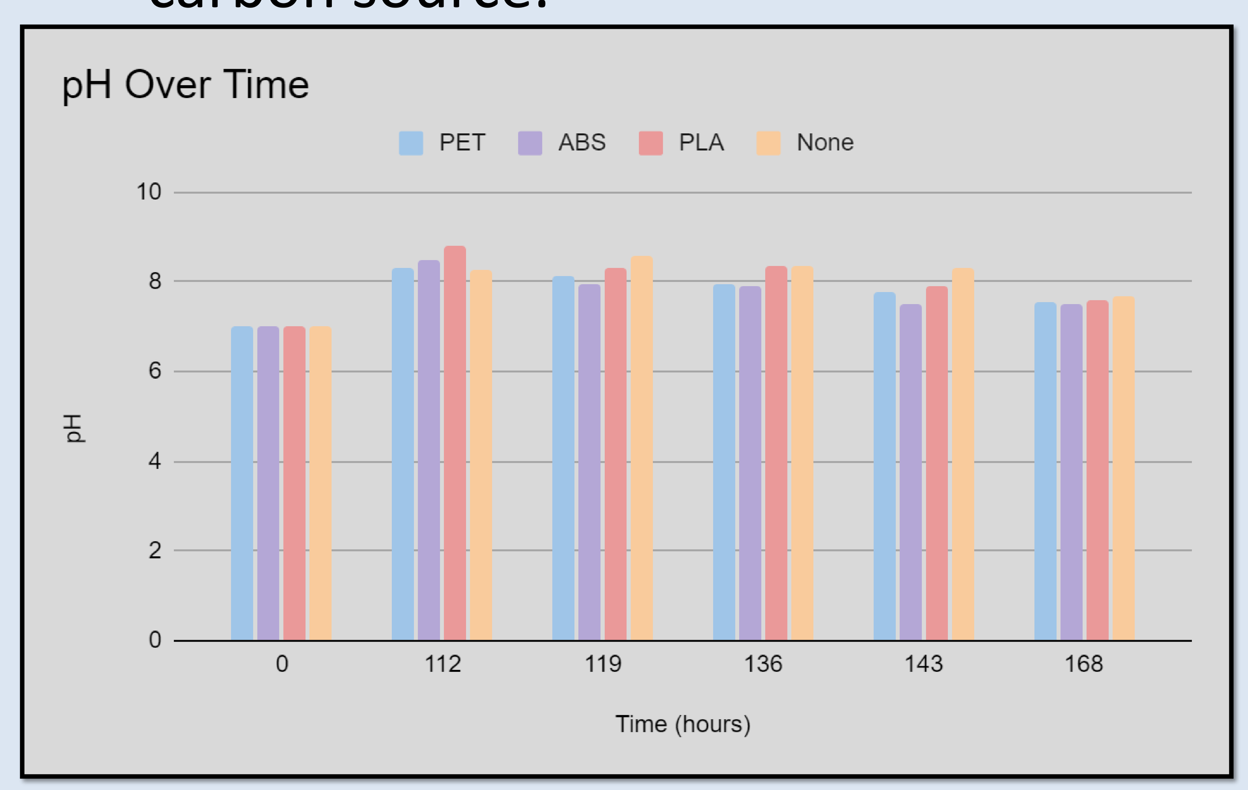
Engineering Goal

Engineer a device to effectively degrade plastic.

Results & Analysis



- Carbon dioxide levels increased for PET and PLA filament groups by an average of 6 ppm, but not for the ABS filament group.
- p-values of 0.91 for all plastics.
- All groups became more basic and then the pH began to decrease.
- The pH of PET was 8.31 112 hours in, and at 168 hours it had decreased to 7.55, however, a similar trend was seen in all groups.
- Mass increased by 0.02 for PET and 0.008 for PLA, and decreased by 0.017 for ABS.
- p-values of 0.98 for PET and PLA, 0.99 for ABS → not significant.
- Final mass was measured to not disrupt the biofilm, so it also measured the mass of the bacteria.
- The increase in mass was likely the bacteria adapting to use the plastic as a carbon source.



Conclusion

- Using bacteria to degrade plastic was not achieved at this time.
- For the first hypothesis that the mass would decrease, it was found that this was not supported since the p-values were between 0.98 and 0.99.
- For the second hypothesis that the carbon dioxide would increase, this was also not significantly supported, with p-values of 0.91 for all three groups.
- For the last hypothesis that the pH would decrease, this was not supported based on the p-values of 0.78, 0.72, and 0.66 for PET, ABS, and PLA respectively.
- Plastic pollution is growing uncontrollably, however, the use of bacteria may allow for a new approach to tackle this issue if a process can be developed.

METHODS

The procedure by Odobašić was followed to make the YSV media and culture the bacteria.

MEDIA

200 mL of the YSV minimal carbon media was placed in a mason jar.

FILAMENT

3 types of plastic filament – PET, PLA, and ABS – were sterilized with ethanol and placed in separate jars with a stir bar.

INCUBATOR

An incubator was used to culture E. coli K12. The stir plate was place in the incubator with the four jars on it and taped shut.

MEASUREMENTS

Mass was measured before and after, and CO_2 and pH readings were taken at regular intervals.

Impact

- A bacteria-oriented recycling process would benefit waste management in both rural and urban areas.
- Discarded plastic could actively be reused in the industry rather than relying on new plastic, thus reducing pollution.
- More land would be open for vegetation as plastic landfills would no longer be needed.
- A process could be developed to separate the ethylene glycol and terephthalic acid from the rest of the solution and media to be able to reuse these substances for plastic.
- In the future, more research could be done to optimize the media for the growth of plastic-degrading bacteria.